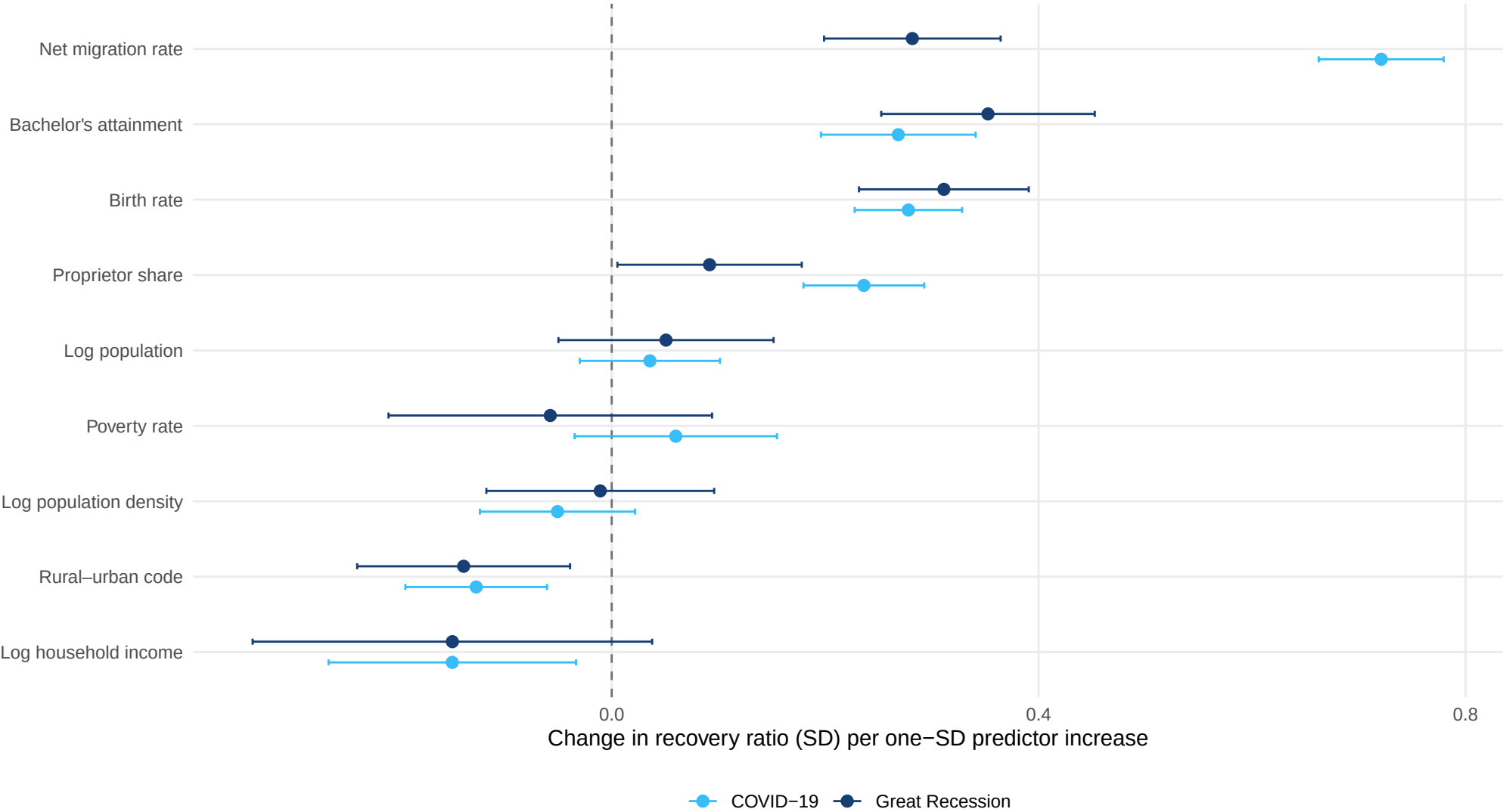


Which pre-crisis characteristics predict recovery?

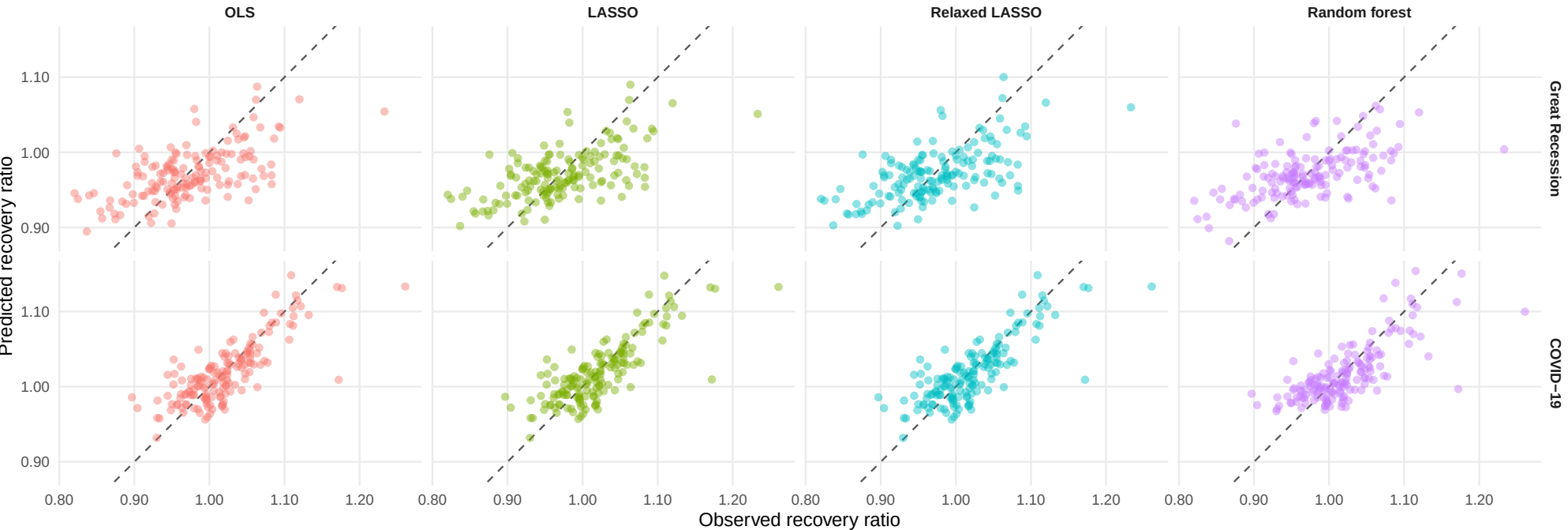
Standardized OLS coefficients with 95% confidence intervals



Target-defining peak, trough, and recovery employment are excluded.

Out-of-sample predictions versus actual recovery

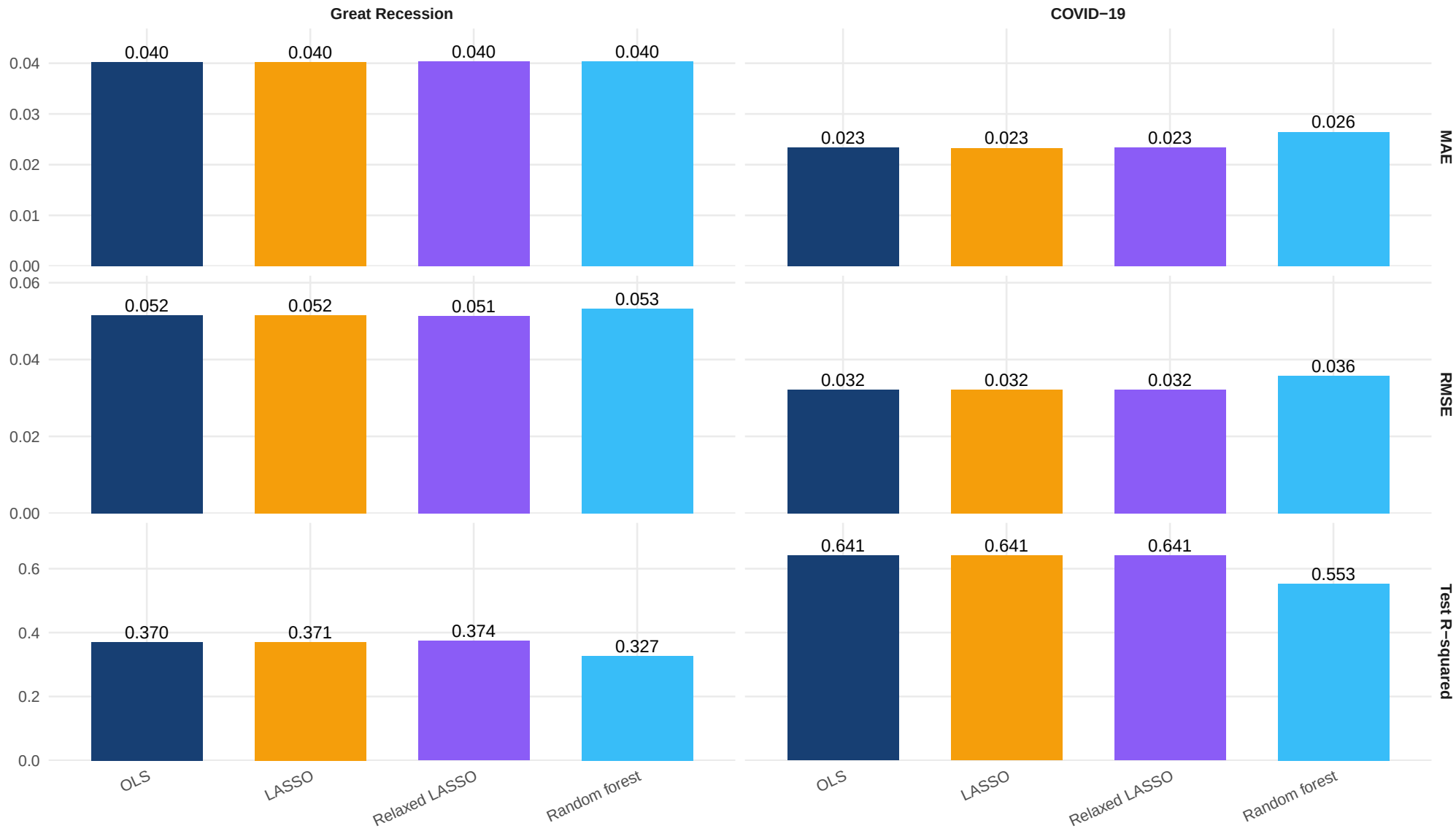
Points closer to the dashed 45-degree line are more accurate



Reproducible 80/20 county split; testing observations only.

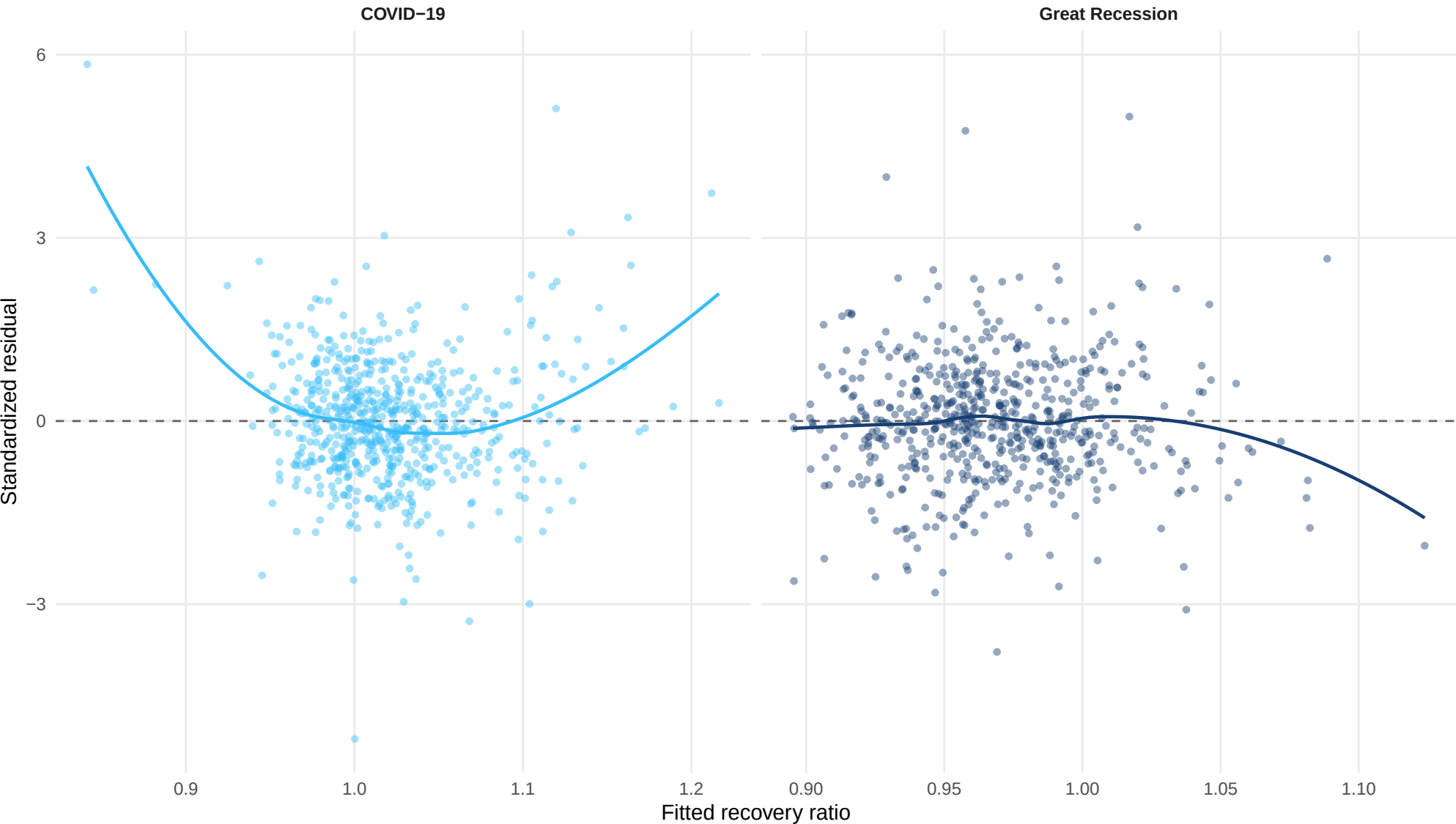
Out-of-sample model performance

Lower RMSE and MAE are better; higher test R-squared is better



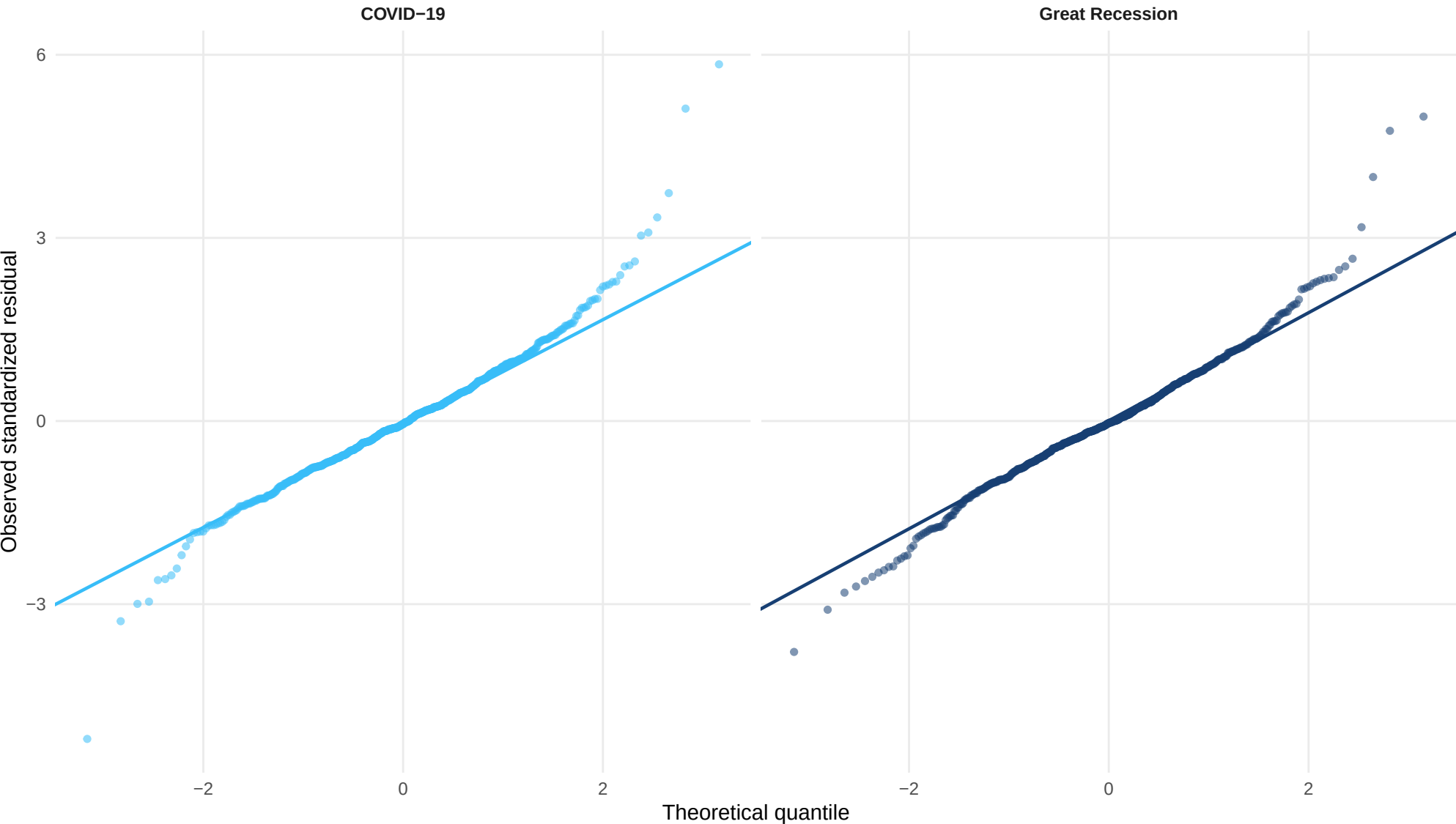
OLS residual diagnostics

A flat smooth line and constant vertical spread support the linear-model assumptions



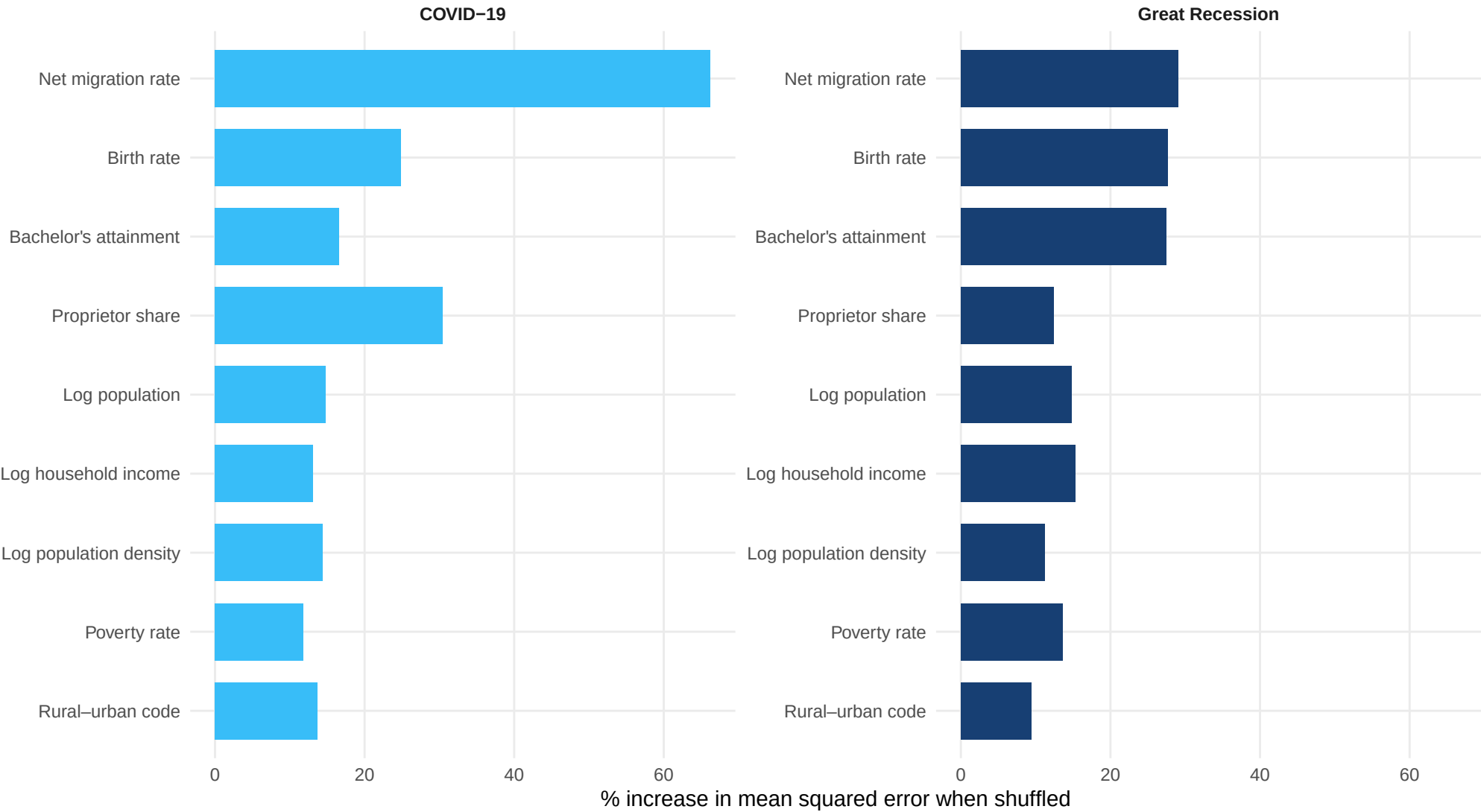
Do the OLS residuals follow a normal distribution?

Systematic tail departures indicate outliers or heavy-tailed errors



Random-forest permutation importance

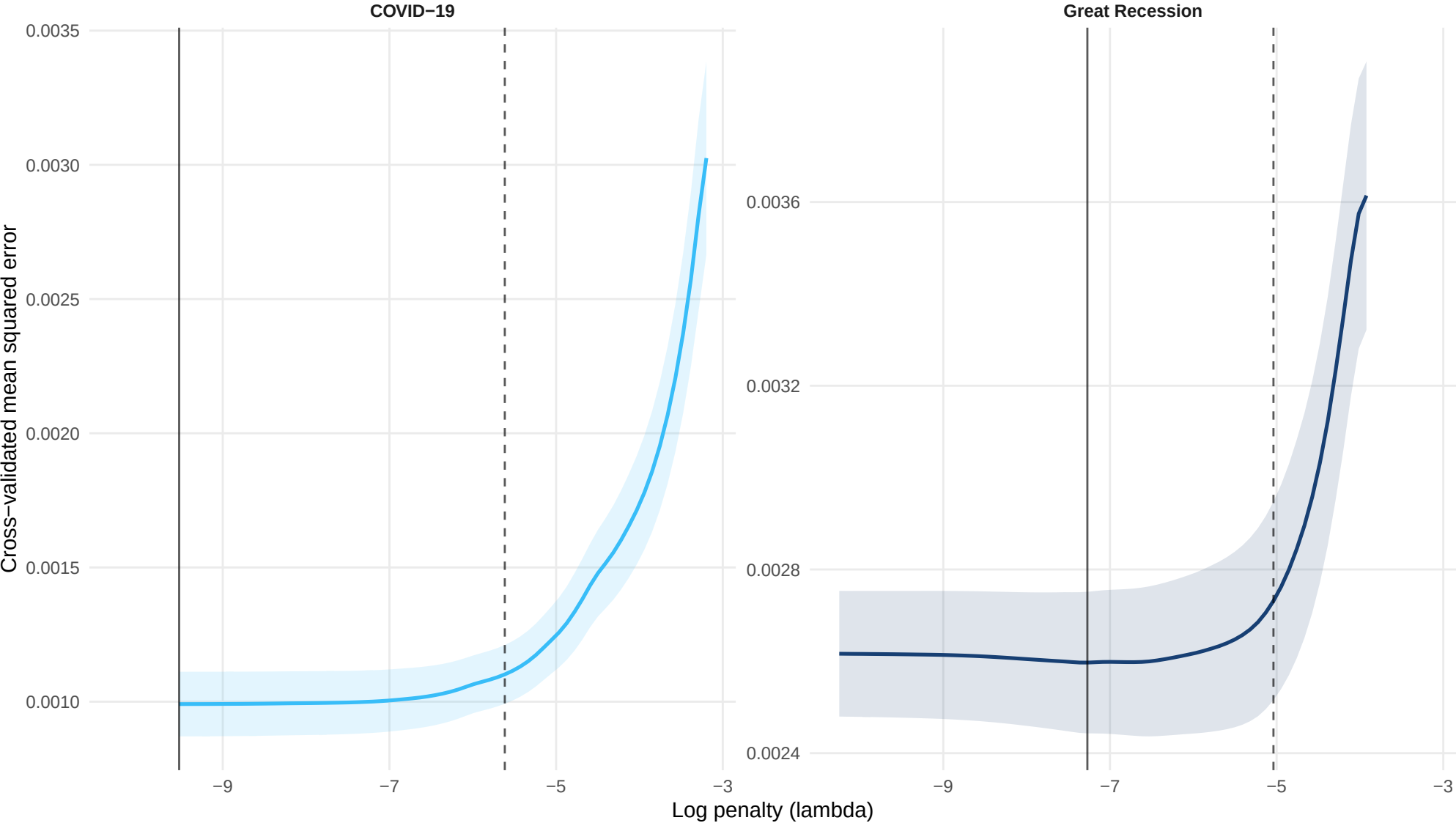
Larger values mean shuffling the predictor caused a larger loss in accuracy



Importance measures predictive reliance—not direction or causality.

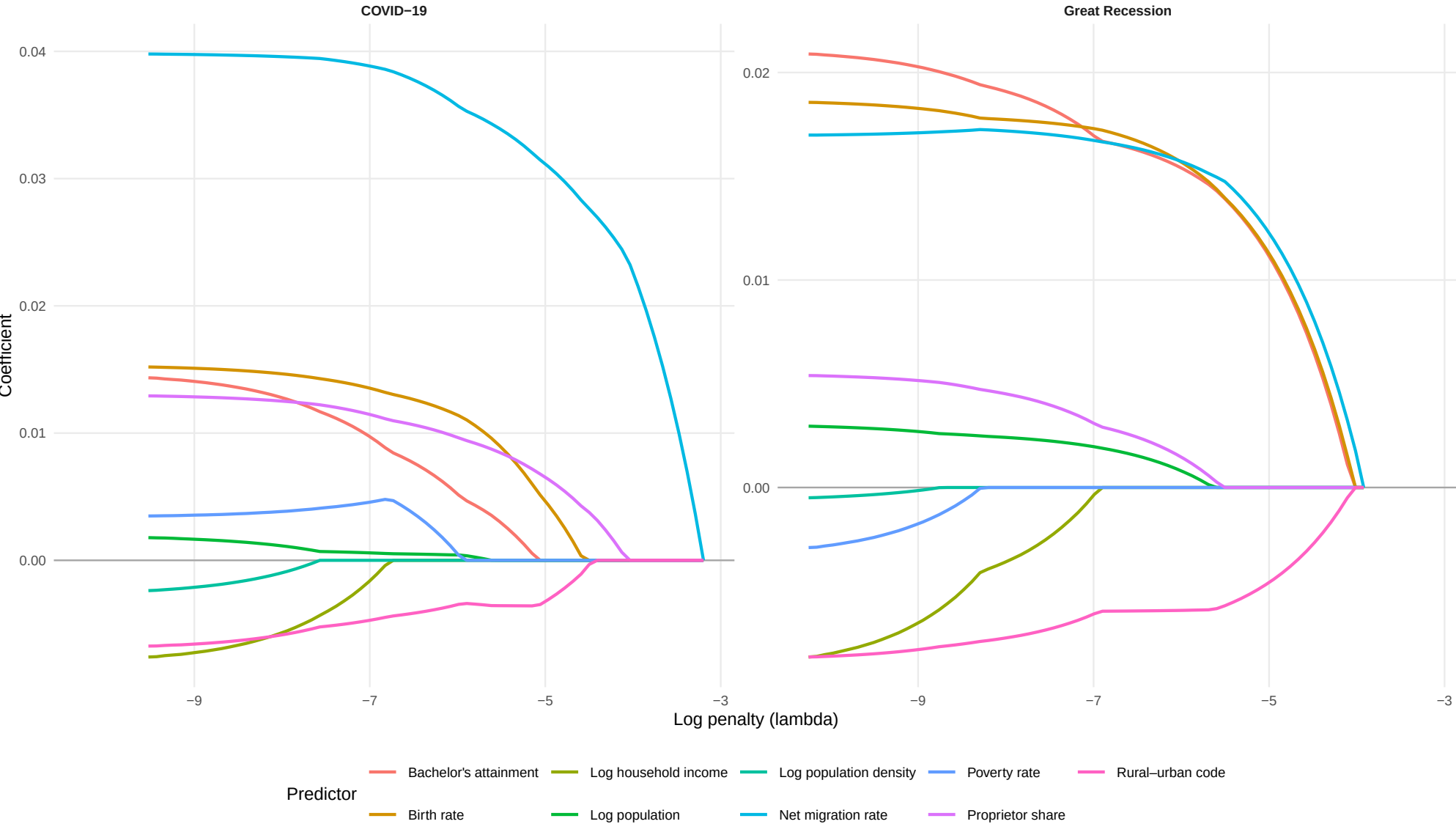
LASSO penalty selected by cross-validation

Solid line: minimum error; dashed line: conservative one-standard-error choice



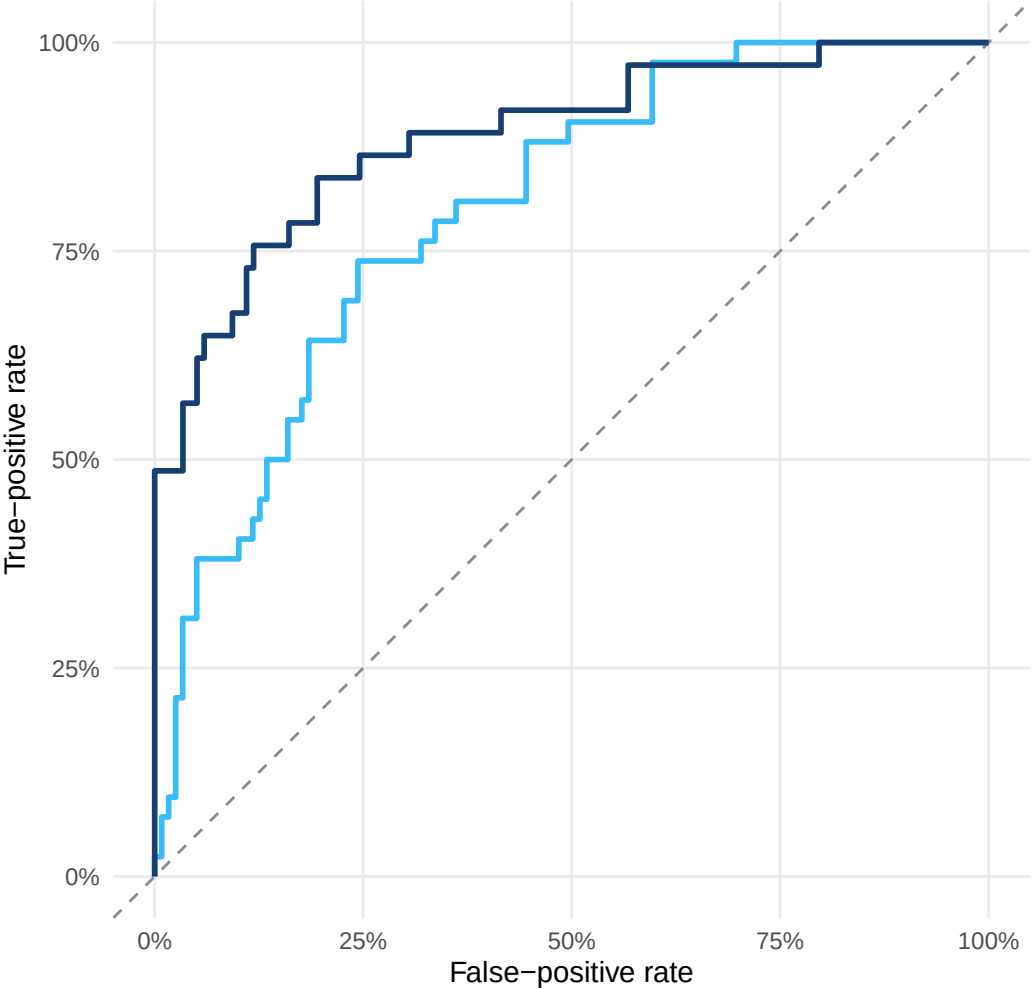
How LASSO shrinks the model

Coefficients move toward zero as the penalty increases



Classifying counties that never recovered

ROC curves for out-of-sample logistic-regression predictions

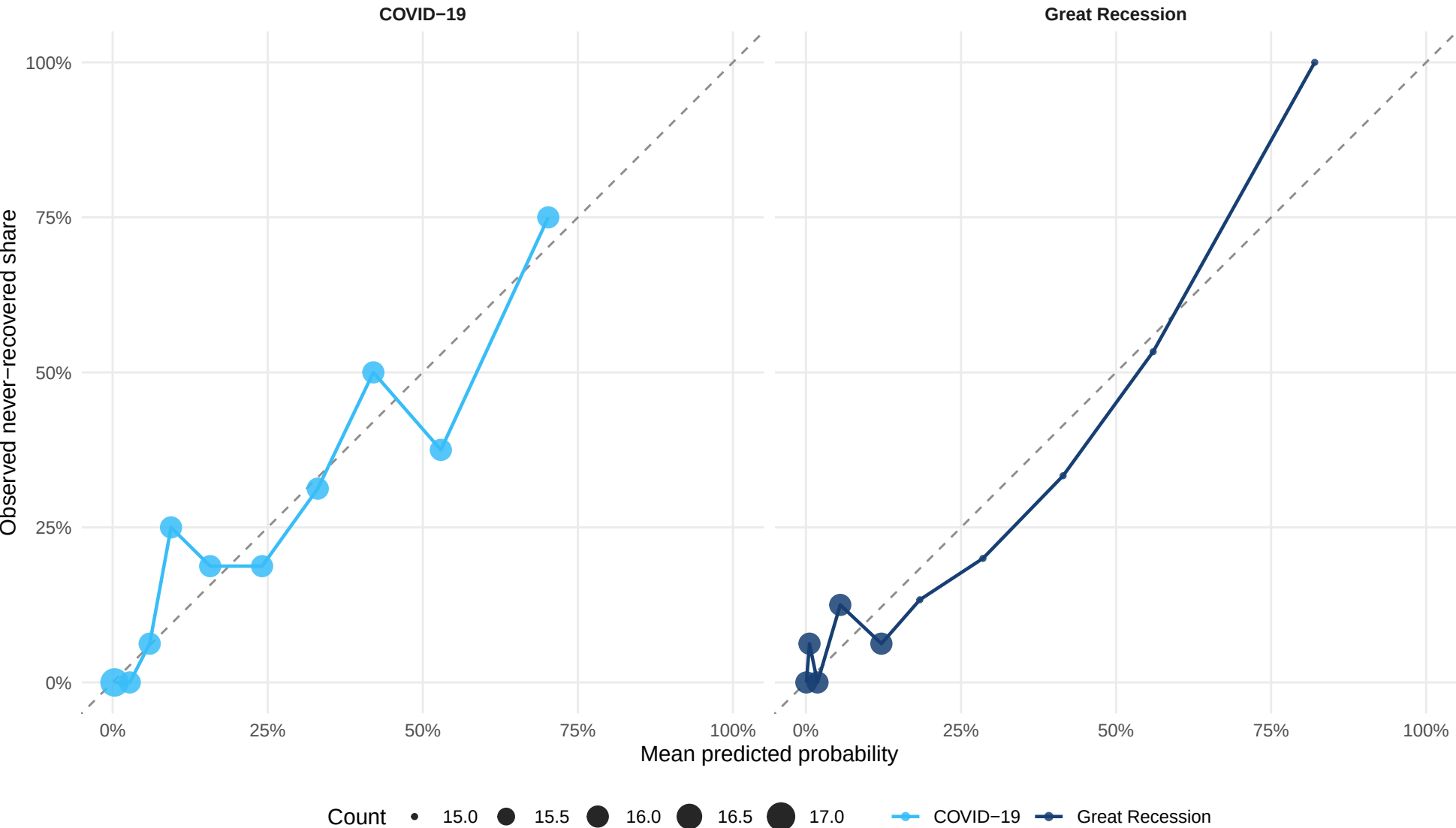


COVID-19 Great Recession

Curves farther above the diagonal indicate better discrimination.

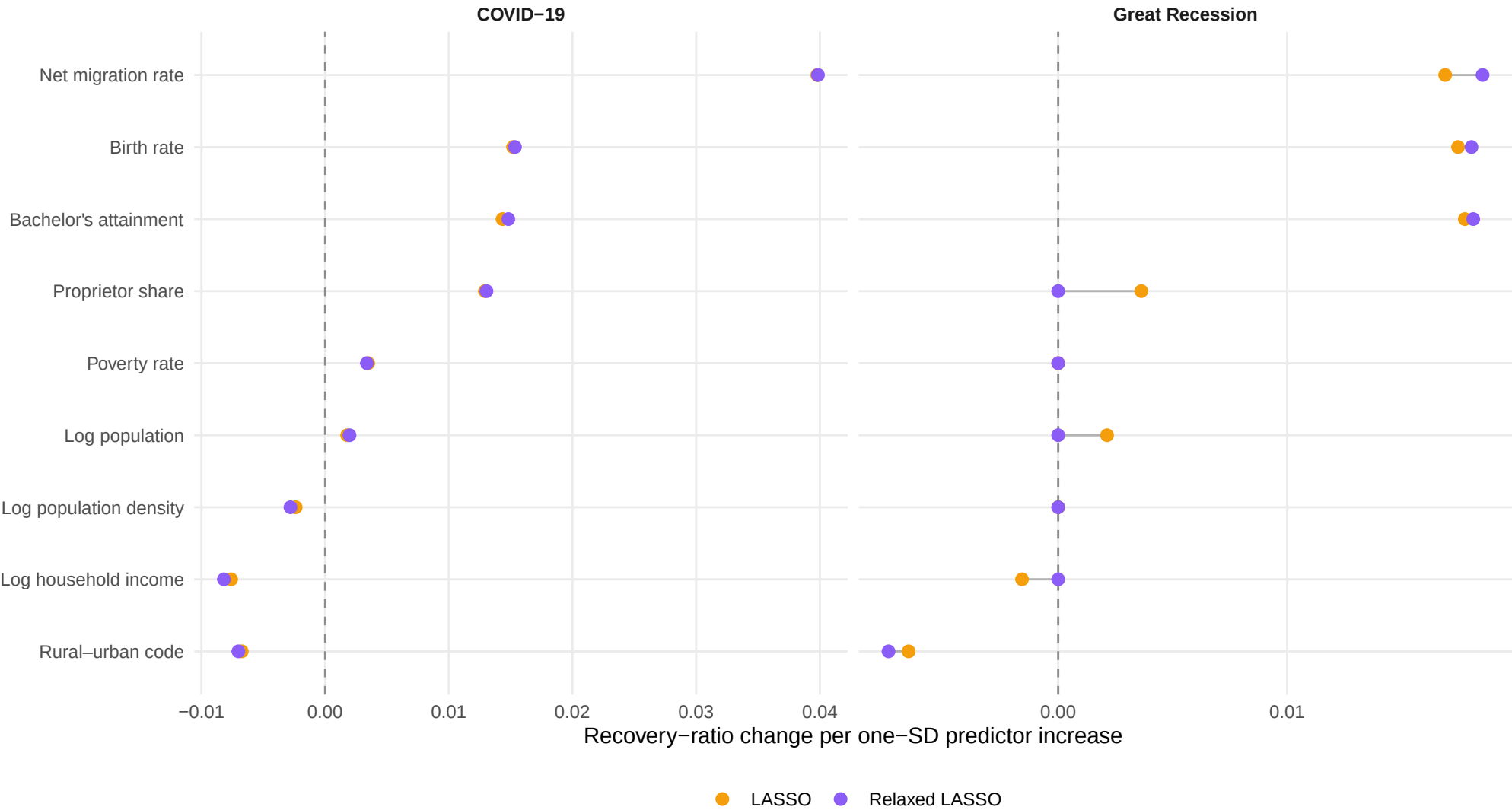
Are the never-recovered probabilities calibrated?

Each point compares predicted and observed rates within a probability decile



How relaxed LASSO changes the selected coefficients

Predictors were standardized before fitting; zeros indicate excluded variables



Relaxation can reduce LASSO shrinkage bias without restoring every predictor.

Relaxed-LASSO tuning

Dashed line marks the cross-validated gamma selected for each crisis

